

Task 2

(1) Why does the initial and final value of the shield power remain unchanged?

(2) Why does the order of output change every time you run the program?

1. The fork() function makes a full copy of the process, including its variables.

Any changes to a variables state in a child process does not affect the parent process. That includes other child processes as well.

2. Parent/child process execution occurs outside of their control. Since they are running as independent processes, the operating system is deciding which running processes get execution time. Sibling processes also don't have any way to synchronize.

Task 4

N=1,000,000

	Sequential	1 processes	2 processes	3 processes	4 processes
Time	real 0m0.655s user 0m0.328s sys 0m0.326s	real 0m0.649s user 0m0.367s sys 0m0.279s	real 0m0.468s user 0m0.376s sys 0m0.202s	real 0m0.451s user 0m0.399s sys 0m0.206s	real 0m0.426s user 0m0.407s sys 0m0.195s

For N=100,000,000, the sequential and single process tests performed with nearly identical results. The multi-process tests yielded very modest speedups. For fun, I turned up NUM_PROCESSES to 16, which is the core count available on my machine in WSL, and there was no noticeable gain at all.

I imagine to get a real performance gains out of this application, the other big chunk, the memory allocation of 100,000,000 elements would need to be parallelized as well. That unit of work is only occurring in the parent process.

I found pipes simple to work with. The problem I would come across was trying to keep track of the idea that a forked process must know which id/integer/file descriptor it's

responsible for at that moment of time. Also, there were instances where I wasn't accumulating the "leftover" elements at the end of the array. i.e. the indexes that weren't evenly distributed between the processes. I had two bugs at two different times: once where the forked process didn't take into account that it was responsible for the extra work, and once where any time NUM_PROCESS wasn't equal to 4. I hard coded the number in my if statement, which caused the bug.